ASSIGNMENT 2

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Question 1: Explain the responsibilities of the Data-Link Layer and distinguish Logical Link Control from Media Access Control.

The Data-Link Layer is Layer 2 of the OSI model. It provides node-to-node delivery of data over a particular network link and organizes network-layer packets into frames.

Responsibilities of the Data-Link Layer

1. Framing: Divides network-layer packets into frames by adding a header and trailer.

2. Physical addressing: Uses MAC addresses to identify network interfaces on a local network.

3. Error detection: Uses a Frame Check Sequence (FCS) to help detect corruption in a received frame.

4. Media access control: Determines how devices share and access a common transmission medium.

5. Flow control: Some data-link protocols help prevent a sender from overwhelming a receiver.

6. Reliable delivery: Some protocols use acknowledgements and retransmissions; this service is not provided in the same way by every data-link protocol.

7. Node-to-node delivery: Transfers frames between devices connected through the same link or LAN.

Logical Link Control (LLC) and Media Access Control (MAC)

Feature	LLC	MAC
Full form	Logical Link Control	Media Access Control
Main purpose	Provides an interface to the network layer and logical-link services	Controls access to the transmission medium
Addressing	Not responsible for ordinary Ethernet MAC addressing	Uses MAC addresses
Main functions	Protocol identification and logical-link services	Framing, physical addressing and medium access
Example	LLC/SNAP protocol identification	Ethernet and Wi-Fi MAC operations

In short, LLC supports communication between the data-link layer and higher-layer protocols, while MAC handles local addressing and access to the physical medium.

Question 2: Explain the Ethernet frame format and identify the purpose of destination MAC, source MAC, type/length and FCS fields.

An Ethernet frame carries data across a local Ethernet network. A standard Ethernet MAC frame is normally 64 to 1518 bytes, from the destination MAC field through the FCS, excluding the preamble and start-frame delimiter.

Ethernet frame format

Preamble 7 bytes	**SFD** 1 byte	**Destination MAC** 6 bytes	**Source MAC** 6 bytes	**Type/Length** 2 bytes	**Data + Padding** 46–1500 bytes	**FCS** 4 bytes

Fields and their purposes

1. Preamble (7 bytes): Helps the receiver synchronize with the incoming transmission.

2. Start Frame Delimiter (SFD, 1 byte): Marks the end of the preamble and the beginning of the frame.

3. Destination MAC address (6 bytes): Identifies the intended destination at the data-link layer. It may identify one device, a group, or all devices on the LAN.

4. Source MAC address (6 bytes): Identifies the transmitting network interface.

5. Type/Length (2 bytes): Values up to 1500 indicate payload length; values of 1536 (0x0600) and above identify the encapsulated protocol. For example, 0x0800 indicates IPv4 and 0x86DD indicates IPv6.

6. Data and padding (46–1500 bytes): Carries the payload, usually a network-layer packet. Padding is added when necessary to meet the minimum frame size.

7. Frame Check Sequence (FCS, 4 bytes): Contains a 32-bit CRC value used to detect corruption. It does not itself correct errors.

Question 3: Use Wireshark to capture Ethernet traffic and identify source and destination MAC addresses from at least five frames.

Wireshark is a packet analyzer that can capture traffic from a selected network interface and display the fields in each packet. For an Ethernet investigation, select the active wired or wireless interface, start capturing, generate traffic (for example, open a website or ping a host), and stop the capture.

Procedure

1. Open Wireshark and select the active network interface.

2. Start the capture and generate some network traffic.

3. Stop the capture after several packets have appeared.

4. Select a packet and expand **Ethernet II** in the packet details pane.

5. Record the source MAC, destination MAC and Type field for at least five frames.

Display filter examples: **eth** shows Ethernet packets, **arp** shows ARP packets, and **icmp** shows ICMP packets.

Example results table

The following table is an illustrative, documented example for explaining how to read the fields. These are not claimed to be a live Wireshark capture.

Frame	Protocol	Source MAC	Destination MAC	Type / EtherType
1	TCP	00:1A:2B:3C:4D:5E	1C:57:DC:1A:2B:3C	0x0800 (IPv4)
2	TCP	1C:57:DC:1A:2B:3C	00:1A:2B:3C:4D:5E	0x0800 (IPv4)
3	ARP	1C:57:DC:1A:2B:3C	FF:FF:FF:FF:FF:FF	0x0806 (ARP)
4	ICMP	00:1A:2B:3C:4D:5E	1C:57:DC:1A:2B:3C	0x0800 (IPv4)
5	ICMP	1C:57:DC:1A:2B:3C	00:1A:2B:3C:4D:5E	0x0800 (IPv4)

Observation

A MAC address identifies a network interface on the local link. The ARP example uses the broadcast destination FF:FF:FF:FF:FF:FF, while the IPv4 examples show unicast MAC addresses. In a submitted practical report, attach readable screenshots from your own capture and ensure the recorded addresses match those screenshots.

Question 4: Explain the operation of a switch using MAC address learning. Demonstrate how a switch builds its forwarding table.

A network switch is a Layer 2 device that forwards Ethernet frames using destination MAC addresses. It learns device locations by examining the source MAC address of each incoming frame.

MAC address learning process

1. A switch receives a frame on a particular port.

2. It reads the source MAC address and records that address with the incoming port.

3. It checks the forwarding table for the destination MAC address.

4. If the destination is known, the switch forwards the frame through the appropriate eligible port.

5. If the destination is unknown, the switch floods the frame through eligible ports in the same VLAN, except the incoming port.

6. Broadcast frames are also forwarded through eligible ports in the same VLAN, except the incoming port.

7. Learned entries can be updated when a device moves, and dynamic entries normally expire after inactivity.

Example network

PC-A MAC AA:AA:AA:AA:AA:AA	**PC-B** MAC BB:BB:BB:BB:BB:BB	**PC-C** MAC CC:CC:CC:CC:CC:CC
Port 1	Port 2	Port 3
Port 1	**Port 2**	**Port 3**
SWITCH	MAC forwarding table	Layer 2 forwarding

Demonstration

1. PC-A sends to PC-B: The switch learns PC-A on Port 1. As PC-B is unknown, the frame is flooded through other eligible ports in the VLAN.

2. PC-B replies to PC-A: The switch learns PC-B on Port 2. Since PC-A is known on Port 1, the reply is forwarded only to Port 1.

3. PC-C sends to PC-A: The switch learns PC-C on Port 3 and forwards the frame to Port 1 because PC-A is already known.

MAC address	Switch port	Learned from
AA:AA:AA:AA:AA:AA	Port 1	Frame received from PC-A
BB:BB:BB:BB:BB:BB	Port 2	Frame received from PC-B
CC:CC:CC:CC:CC:CC	Port 3	Frame received from PC-C

Thus, a switch builds its forwarding table dynamically by associating source MAC addresses with incoming ports. This reduces unnecessary traffic compared with repeating every frame to every port.

Question 5: Compare hub, switch and bridge in terms of forwarding, collision domains and practical use.

Hubs, bridges and switches connect devices or LAN segments, but they differ in the way they handle frames and divide network traffic.

Hub

A hub operates at the Physical Layer (Layer 1). It repeats incoming signals to the other ports and does not inspect destination MAC addresses. All connected devices share one collision domain, so simultaneous transmissions may collide in half-duplex Ethernet.

Bridge

A bridge operates at the Data-Link Layer (Layer 2). It learns source MAC addresses and filters or forwards frames between LAN segments. It separates collision domains but generally does not separate broadcast domains unless Layer 3 or VLAN functionality is also used.

Switch

A switch is effectively a multiport bridge. It learns MAC addresses and forwards known unicast frames only through the appropriate port. Each switch port is a separate collision domain; in full-duplex switched Ethernet, collisions do not occur during normal operation.

Comparison table

Feature	Hub	Bridge	Switch
OSI layer	Physical (Layer 1)	Data-Link (Layer 2)	Data-Link (Layer 2)
Forwarding	Repeats signals to other ports	Filters/forwards frames using MAC addresses	Uses a MAC table to forward frames
MAC learning	No	Yes	Yes
Collision domains	One shared collision domain	Separates collision domains by segment	Each port is a separate collision domain
Bandwidth	Shared by connected devices	Depends on bridge architecture and links	Dedicated port bandwidth in switched full-duplex Ethernet
Duplex	Typically half-duplex	Depends on implementation	Commonly full-duplex
Practical use	Older, largely obsolete LANs	Connecting or segmenting LAN sections	Modern wired LANs

Question 6: Explain CSMA/CD and discuss why its importance is reduced in modern switched full-duplex Ethernet.

CSMA/CD stands for Carrier Sense Multiple Access with Collision Detection. It was used in traditional shared, half-duplex Ethernet to manage access to a common medium.

Operation of CSMA/CD

1. Carrier sensing: A device listens to check whether another device is transmitting.

2. Multiple access: Several devices share the same transmission medium.

3. Transmission: If the medium is idle, the device begins transmitting.

4. Collision detection: While transmitting, the device monitors the medium for a collision.

5. Collision response: If a collision is detected, the device stops and sends a jam signal.

6. Backoff: It waits for a random interval determined by binary exponential backoff.

7. Retransmission: It attempts to transmit again after the waiting period.

Why CSMA/CD is less important today

• Modern Ethernet commonly uses switches with dedicated full-duplex links to devices.

• A full-duplex link supports transmitting and receiving simultaneously.

• The two directions do not compete on a shared half-duplex medium.

• There is no shared collision domain on a normal full-duplex link, so collisions do not occur during normal operation.

• Therefore, CSMA/CD and its collision-recovery procedure are not required on modern switched full-duplex links.

CSMA/CD was important in shared half-duplex Ethernet, particularly networks built with hubs. Switched full-duplex Ethernet largely removes the collision problem.

Question 7: Explain CSMA/CA in IEEE 802.11 and compare its objective with CSMA/CD.

CSMA/CA stands for Carrier Sense Multiple Access with Collision Avoidance. It is used in IEEE 802.11 wireless LANs to reduce the chance of simultaneous transmissions. Wireless stations cannot reliably detect collisions while transmitting in the same way traditional wired Ethernet devices can.

Operation of CSMA/CA

1. Carrier sensing: A station checks whether the wireless channel is busy.

2. Waiting: If the channel is busy, the station waits until it becomes available.

3. Interframe spacing: The station waits for the required interframe interval.

4. Random backoff: It selects a random counter from a contention window and decreases it while the channel remains idle.

5. Transmission: When the counter reaches zero, the station transmits the frame.

6. Acknowledgement: The receiver normally acknowledges successful reception.

7. Retransmission: If no acknowledgement arrives, the sender treats delivery as possibly unsuccessful, increases contention as required and retries, subject to retry limits.

RTS/CTS mechanism

IEEE 802.11 may use Request to Send (RTS) and Clear to Send (CTS) frames to reserve the medium before sending data. This can help with the hidden-node problem, where two stations cannot hear each other but can both communicate with the same access point. RTS/CTS is optional and is not used for every frame.

CSMA/CD and CSMA/CA

Feature	CSMA/CD	CSMA/CA
Full form	Carrier Sense Multiple Access with Collision Detection	Carrier Sense Multiple Access with Collision Avoidance
Main use	Traditional shared half-duplex Ethernet	IEEE 802.11 wireless LANs
Objective	Detect collisions after they occur	Reduce the probability of collisions
Medium	Shared wired Ethernet medium	Shared wireless radio channel
Collision handling	Detect, stop, back off and retry	Contend for the channel; infer failure using acknowledgements and retry
Acknowledgements	Not normally used for ordinary Ethernet frame delivery	MAC acknowledgements are commonly used
RTS/CTS	Not part of ordinary Ethernet CSMA/CD	May be used to reserve the wireless medium

Both mechanisms coordinate access to a shared medium, but CSMA/CD reacts to detected collisions whereas CSMA/CA attempts to avoid them before transmission.

Question 8: Design a small wireless LAN for a college laboratory. Specify access point placement, SSID, channel considerations, addressing and basic security requirements.

The goal is to provide reliable wireless connectivity for student laptops, an instructor device and a network printer while reducing interference and protecting the laboratory network.

1. Access point placement

Place the access point near the centre of the laboratory in an elevated, open position. Avoid metal cabinets and major sources of interference. Conduct a site survey and add managed access points if coverage or capacity is insufficient.

2. SSID

Use **College-Lab** as the SSID. If required, configure separate staff and guest SSIDs with different access permissions.

3. Channel considerations

• For 2.4 GHz, use 20 MHz channel width and select a channel based on a local survey; channels 1, 6 and 11 are commonly used as non-overlapping choices where permitted.

• For 5 GHz, select a permitted channel with less interference.

• Consider 6 GHz only if supported by the equipment and allowed by local regulations.

4. IP addressing plan

Device	Example address / configuration
Default gateway	192.168.10.1
Access point management	192.168.10.2
Wi-Fi printer	192.168.10.10
Student laptops	DHCP range: 192.168.10.100–192.168.10.199
Instructor laptop	DHCP reservation or suitable assigned address
Subnet mask	255.255.255.0 (/24)

DHCP automatically assigns addresses to student devices and reduces manual configuration. Keep infrastructure addresses outside the dynamic pool or reserve them to avoid address conflicts.

5. Basic security requirements

• Use WPA3-Personal where supported, or WPA2-AES for compatibility. For a managed college network, WPA2/WPA3-Enterprise with 802.1X and RADIUS is preferable when available.

• Use a strong, unique wireless password and change default administrator credentials.

• Disable WPS if it is not needed and update access-point firmware regularly.

• Separate student, staff and guest access with VLANs or separate networks where possible.

• Enable guest client isolation when appropriate and restrict access to the access-point management interface.

• Review connected devices and logs periodically.

Expected outcome: suitable access-point placement improves coverage, channel planning reduces interference, DHCP simplifies addressing, and security controls help protect the network.

Question 9: Explain Mobile IP and the need for mobility support when a device moves between networks.

Mobile IP is a networking concept designed to let a mobile device move between IP networks while maintaining a stable home IP address for communication. It aims to support mobility without requiring every communicating device to know the mobile device's current network location.

Ordinary IP routing relies on addresses associated with network location. When a device moves to another network, its point of attachment changes. If it obtains a new IP address, ongoing sessions may be disrupted unless mobility support is available.

Important terms

Term	Meaning
Mobile Node (MN)	The device that moves between networks.
Home Network	The network associated with the mobile node's permanent home address.
Home Address	The stable IP address assigned to the mobile node in its home network.
Home Agent (HA)	In Mobile IPv4, a router on the home network that tracks the mobile node's current location and can tunnel packets to it.
Care-of Address (CoA)	An address representing the mobile node's current point of attachment or reachable location.
Foreign Network	A network other than the mobile node's home network.
Correspondent Node	A device communicating with the mobile node.

Working of Mobile IPv4

1. The mobile node starts in its home network and communicates using its home address.

2. When it moves to another network, it obtains or discovers a care-of address.

3. The mobile node registers the care-of address with its home agent.

4. A correspondent node sends packets to the mobile node's home address.

5. The home agent intercepts relevant packets and tunnels them to the care-of address.

6. The mobile node receives the packets through the new network.

7. Depending on the routing arrangement, replies may travel directly to the correspondent node or through a tunnel.

Need for mobility support

• **Continuity:** Helps maintain communication while a device changes networks.

• **Stable identity:** Allows a device to retain its home address while its attachment point changes.

• **Session support:** Can reduce disruption to ongoing applications.

• **Location management:** Maintains information about the mobile node's current reachability.

• **Roaming:** Supports movement between networks without relying only on manual reconfiguration by applications.

Question 10: Perform a packet-level comparison of a wired Ethernet frame and a wireless LAN frame using suitable captures or documented examples.

Ethernet is specified by IEEE 802.3, while Wi-Fi MAC framing is specified by IEEE 802.11. Both use MAC addressing and a Frame Check Sequence, but their header structures and medium-access behavior differ.

Wired Ethernet frame

Destination MAC 6 bytes	**Source MAC** 6 bytes	**Type/Length** 2 bytes	**Data** 46–1500 bytes	**FCS** 4 bytes

Wireless LAN frame (basic three-address 802.11 data frame)

Frame Control 2 bytes	**Duration/ID** 2 bytes	**Address 1** 6 bytes	**Address 2** 6 bytes
Address 3 6 bytes	**Sequence Control** 2 bytes	**Frame Body / Payload**	**FCS** 4 bytes

Packet-level comparison

Feature	Wired Ethernet	IEEE 802.11 Wireless LAN
Standard	IEEE 802.3	IEEE 802.11
Medium	Wired Ethernet link	Wireless radio channel
Basic addressing	Destination and source MAC addresses	Address fields whose meaning depends on frame type and direction
Header	Typically 14 bytes without VLAN tagging, excluding FCS	Commonly 24 bytes for a basic three-address data frame, before optional fields
Address fields	Destination MAC and source MAC	May identify receiver, transmitter, source, destination and BSSID
Frame control	No equivalent Frame Control field in the basic Ethernet header	Identifies frame type, subtype and control flags
Sequence control	No corresponding general-purpose field in the basic Ethernet header	Carries sequence and fragmentation information in relevant frames
Error detection	32-bit FCS	32-bit FCS in ordinary 802.11 MAC frames
Access method	Modern switched full-duplex Ethernet does not use CSMA/CD	CSMA/CA is used for contention-based channel access
Acknowledgement	Ordinary Ethernet does not normally acknowledge every frame at MAC level	Unicast frames are commonly acknowledged at MAC level, subject to frame type and mode

Packet inspection notes

1. In Wireshark, an Ethernet capture commonly displays an Ethernet II header.

2. A Wi-Fi capture in monitor mode may show the original 802.11 MAC header; a normal Wi-Fi interface may instead show decapsulated traffic.

3. Compare the source and destination addresses, header fields, frame length and protocol information visible in the packet details pane.

4. Attach screenshots of actual captures if required by the instructor. Do not label illustrative examples as live captures.

CONCLUSION

The Data-Link Layer enables local frame delivery through framing, MAC addressing, error detection and medium-access rules. Ethernet switches improve forwarding efficiency by learning MAC addresses, while full-duplex switched links remove the need for collision detection. Wireless LANs use CSMA/CA because multiple stations share radio spectrum, and a well-designed laboratory WLAN requires sensible access-point placement, channel planning, addressing and security. Mobile IP illustrates how mobility mechanisms can help preserve communication when a device changes networks. Finally, Ethernet and 802.11 frames share some functions but use different header structures and access methods.

References

• IEEE 802.3 Ethernet standard and IEEE 802.11 Wireless LAN standard (standard family references).

• Wireshark User's Guide, packet capture and display-filter documentation.

• Course material and the provided assignment reference document.

README.md — submission checklist

■ PDF report with answers to all ten questions

■ Ethernet frame diagram

■ Wireshark screenshots from the actual capture, if required

■ Wireless LAN design and IP addressing plan

■ Hub/bridge/switch and Ethernet/Wi-Fi comparison tables

■ References for any external material used

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